

STA130 Project

Project subtitle (if applicable)

Yifa , Lucas , Nitalia , Linus | Group #X

2026-03-17

Overall Theme: [Factors associated with electricity access and performance in African countries.]

Research Goal: [The research goal is to examine whether GDP per capita is a significant predictor of electricity access rates across African countries.]

Target Population: [Yifan , Lucas , Natalia , Linus]

Audience: [Who is this for?]

Key Variables:

- Variable 1: [GDP per capita (measure of economic development for each country)]
- Variable 2: [Electricity access rate (percentage of population with access to electricity)]

Data Wrangling:

- [Aggregated the dataset to one observation per country by taking averages
Removed missing values for GDP and electricity access rate
Created GDP groups (high vs low) based on the median GDP per capita]

Question #1: [Yifan - Is GDP per capita a significant predictor of electricity access rates across African countries]

[Is GDP per capita a significant predictor of electricity access rates across African countries?

This analysis examines whether countries with higher levels of economic development tend to have better access to electricity.]

Q1: Variables & Approach

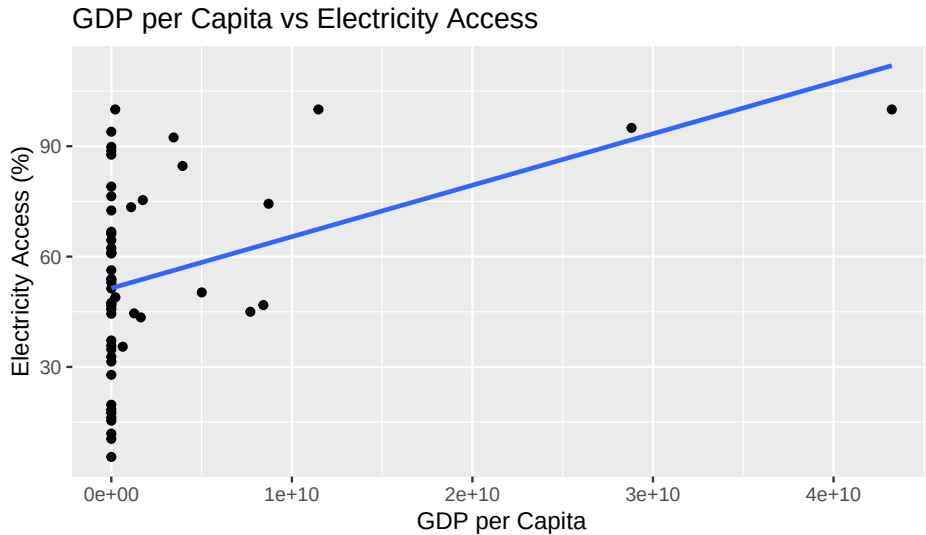
Variables used:

- [- GDP per capita ('gdp')
 - Electricity access rate ('electrification_rate')
 - Country]

New variables/groups defined:

- [- Aggregated data to one observation per country by taking averages
 - Created GDP groups (high vs low) based on median GDP]

Q1: Visualization



Q1: Summary Table

```
# A tibble: 2 x 2
  gdp_group avg_access
  <chr>      <dbl>
1 High GDP    67.9
2 Low GDP     41.6
```

Insights for Q1

Question #2: [Nitalia - Question Statement]

[How large is the gap between urban and rural electricity access across African countries?]

Q2: Variables & Approach

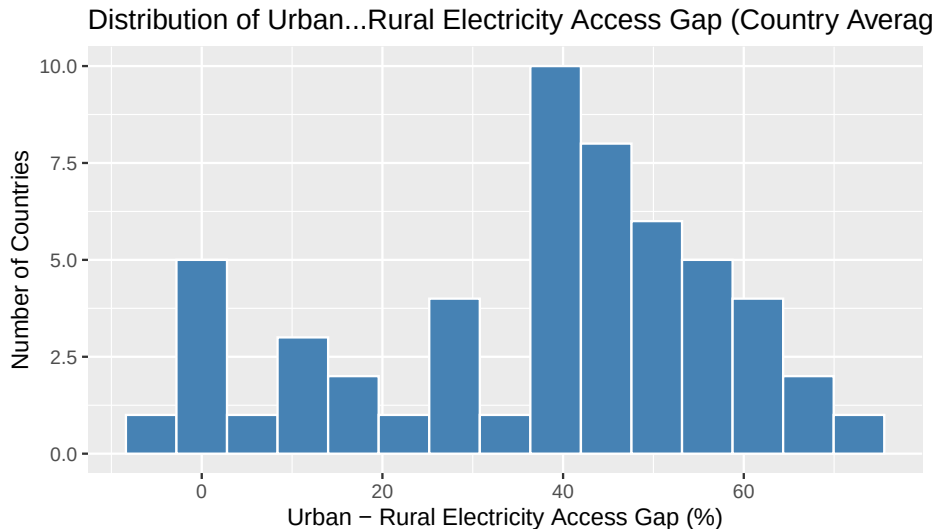
Variables used:

- `urban_access_electricity_rate` - percentage of urban population with electricity access
- `rural_access_electricity_rate` - percentage of rural population with electricity access
- `country`

New variables defined

- `access_gap` = `urban_access_electricity_rate` - `rural_access_electricity_rate`
 - measures the inequality between urban and rural electricity access.

Q2: Visualization



Q2: Summary Table

Table 1: Urban and Rural Electricity Access (Country Averages)

Countries	Average Urban Access (%)	Average Rural Access (%)	Average Access Gap (%)	Largest Gap (%)
54	75.9	38.6	37.4	75.2

```
# A tibble: 1 x 4
  country      avg_urban_access avg_rural_access access_gap
  <chr>          <dbl>          <dbl>      <dbl>
1 South Africa      90.9             94      -3.10
```

Table 2: Summary of Urban and Rural Electricity Access Across African Countries

Countries	Average Urban Access (%)	Average Rural Access (%)	Average Access Gap (%)	Largest Gap (%)
585	75.3	41.1	34.3	90.2

Question #3: [Lucas - What is the distribution of electricity outage duration (SAIDI) across African countries?]

This question examines how electricity outage duration varies across African countries. SAIDI measures the total outage time experienced over a year, so it is a direct indicator of electricity reliability. Countries with higher SAIDI values experience longer total outage duration, which signals weaker electricity system performance.

Q3: Variables & Approach

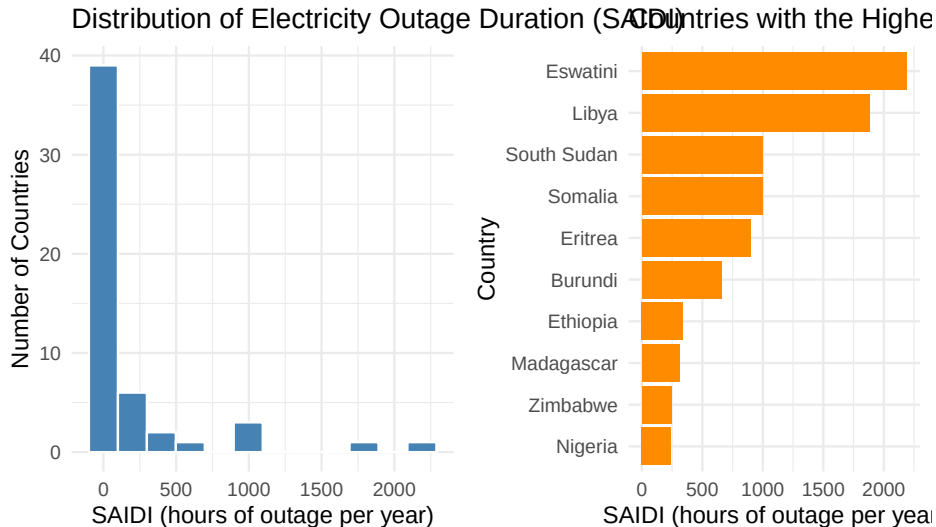
Variables used:

- country
- SAIDI

Approach:

- Use the cleaned country-level dataset CATF_clean
- Remove countries with missing SAIDI values
- Show the overall distribution of SAIDI using a histogram
- Show the countries with the highest SAIDI values using a bar chart

Q3: Visualization



Q3: Summary Table

Table 3: Summary of SAIDI Across African Countries

Countries	Median SAIDI	Mean SAIDI	IQR SAIDI	Minimum SAIDI	Maximum SAIDI
53	41.7	194.1	91.3	0	2190

Insights for Q3

- SAIDI varies substantially across African countries, showing that electricity reliability is uneven across the region.
- The histogram shows whether most countries have relatively low outage duration or whether the distribution is pulled upward by a smaller number of high-outage countries.
- The bar chart identifies the countries with the highest outage burden, which helps highlight where reliability problems appear most severe.

Question #4: [Name - Question Statement] (if applicable)

Q4: Variables & Approach

Variables used:

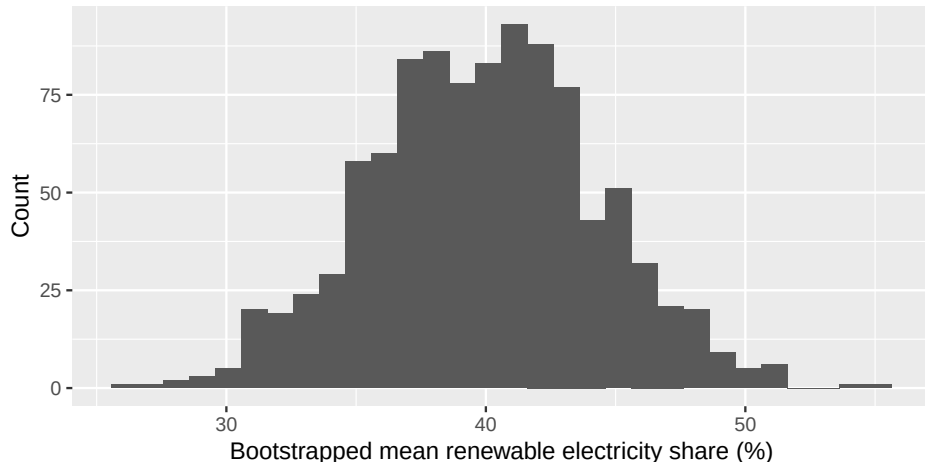
- Renewable electricity generation share (%)
- Country

New variables/groups defined:

- A cleaned renewable share variable with missing values removed
- A bootstrap sample mean for each resample
- A 95% bootstrap confidence interval for the average renewable share

Q4: Visualization

Bootstrap Distribution of Mean Renewable Electricity Share



Based on 1000 bootstrap resamples of African countries.

The bootstrap distribution is centered around approximately 40%, suggesting that the

Q4: Summary Table

Table 4: Summary of Renewable Electricity Share Across African Countries

Number of countries	Sample mean (%)	Sample median (%)	Standard deviation (%)	95% CI lower bound (%)	95% CI upper bound (%)
54	39.8	31	32.1	31.4	48.5

The summary table shows that the average renewable electricity share is approximately 39.8%, with a 95% confidence interval ranging from about 31.4% to 48.5%. This interval suggests that while the average renewable share is around 40%, there is some uncertainty in the estimate. The relatively wide range reflects variation in renewable electricity usage across countries.

Overall Limitations

It is optional to include this as a stand-alone section at the end, you could also choose to discuss this at the end of each research question

Data Limitations:

- [The dataset contains missing values for some countries, which may reduce the representativeness of the analysis]
- [The data is aggregated at the country level, which may hide variation within countries (e.g., urban vs rural differences)]

Future Directions:

- [Future analysis could include additional variables such as infrastructure development, policy factors, and geographic conditions to better explain differences in electricity access]

Overall Conclusion

Key Findings:

- [GDP per capita is positively associated with electricity access across African countries]
- [Countries with higher GDP per capita have significantly higher electricity access rates]
- [High-GDP countries have about 26 percentage points higher electricity access than low-GDP countries]

Connection to Theme:

[The findings highlight the importance of economic development in improving electricity access, which is a key component of energy sustainability and infrastructure development across African countries]

References & Acknowledgements (if applicable)

References:

- [Citation 1, if applicable]

Acknowledgements:

- [Any acknowledgements, if applicable]